Rational Inattention Model for Scoring Rules

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1 The rational inattention framework

1.1 Core idea

Rational inattention (Sims, 2003) models agents who have limited information-processing capacity. The central premise is that extracting, processing, or acting on information is cognitively costly. Rather than assuming agents make random errors, rational inattention treats the *precision* of an agent's behavior as a choice variable: the agent optimally allocates scarce cognitive resources, paying more attention to decisions where the stakes are high and less where they are low.

The framework was originally developed for information acquisition: an agent facing uncertainty about the state of the world chooses how much information to gather, subject to an information-processing cost measured by mutual information (or equivalently, the expected KL divergence between posterior and prior beliefs). Matějka & McKay (2015) applied this to discrete choice problems—settings where an agent chooses among finitely many alternatives (e.g., which product to buy, which route to take). They showed that a rationally inattentive agent's choice probabilities take the *multinomial logit* form: the probability of choosing alternative j is proportional to $\exp(u_j/c)$, where u_j is the payoff and $c > 0$ is the information cost parameter—higher c means information is more expensive and choices are less sensitive to payoffs. The multinomial logit is one of the most widely used models in empirical economics (demand estimation, transportation, industrial organization), but had previously been motivated only by assuming that payoff shocks follow a particular extreme-value distribution—an assumption with no behavioral justification. Matějka & McKay showed that the same model arises from optimal information processing under cognitive constraints, replacing the arbitrary distributional assumption with a decision-theoretic foundation.

1.2 Application to forecasting: information acquisition

We apply the rational inattention framework in its original sense: the agent faces *uncertainty about the state* and chooses how much information to acquire before acting. This is the natural interpretation for forecasting—a forecaster does not know the true probability p and must investigate, deliberate, and gather evidence to form an estimate. The scoring rule determines how valuable this information is.

Formally, the true probability p is drawn from a prior $f(p)$ shared by the principal and the agent. The agent chooses an *information strategy*: a joint distribution over states p and reports $\hat{p}$, subject to the constraint that the marginal over p equals the prior f. Equivalently, the agent chooses a family of conditional reporting distributions $\pi(\hat{p} \mid p)$. The agent's problem is:

$$\max_{\pi(\hat{p}|p)} \left\{ \int_0^1 f(p) \int_0^1 \pi(\hat{p} \mid p)\, V_p(\hat{p})\, d\hat{p}\, dp \;-\; c \cdot I(p; \hat{p}) \right\}, \tag{1}$$

where:

- $V_p(\hat{p}) = p\,S(\hat{p},1) + (1-p)\,S(\hat{p},0)$ is the expected score from reporting $\hat{p}$ when the true probability is p,
- $c > 0$ is the *information cost parameter*, measuring the cognitive cost per unit of information acquired,
- $I(p;\,\hat{p})$ is the mutual information between the state and the report, defined as the expected KL divergence of the conditional reporting distribution from the marginal:

$$I(p;\,\hat{p}) = \int_0^1 f(p)\,D_{\mathrm{KL}}\big(\pi(\hat{p}\mid p)\,\big\|\,q(\hat{p})\big)\,dp,$$

  where $q(\hat{p}) = \int_0^1 f(p)\,\pi(\hat{p}\mid p)\,dp$ is the *marginal distribution of reports* (the unconditional probability of reporting $\hat{p}$, averaged over states), and the KL divergence between two distributions π and q is:

$$D_{\mathrm{KL}}(\pi(\hat{p}\mid p)\|q) = \int_0^1 \pi(\hat{p}\mid p)\ln\frac{\pi(\hat{p}\mid p)}{q(\hat{p})}\,d\hat{p}.$$

The first term is the expected payoff from the chosen information strategy. The second term is the cost of information: acquiring more precise signals about p requires more cognitive effort, and the mutual information $I(p;\,\hat{p})$ measures how much the agent's report reveals about the true state. When $c = 0$, the agent acquires perfect information and reports $\hat{p} = p$. As $c \to \infty$, information is prohibitively expensive and the agent's report becomes independent of p—they fall back on some fixed default distribution q_0.

1.3 The endogenous reference distribution

A key feature of the information acquisition formulation (1) is that the reference distribution is *endogenous*. The mutual information $I(p;\,\hat{p})$ can be decomposed as the expected KL divergence of the conditional reporting distribution $\pi(\hat{p}\mid p)$ from the marginal $q(\hat{p})$:

$$I(p;\,\hat{p}) = \int_0^1 f(p)\,D_{\mathrm{KL}}\big(\pi(\hat{p}\mid p)\,\big\|\,q\big)\,dp.$$

The marginal $q(\hat{p})$ plays the role of the reference distribution, but it is not chosen by the modeler—it is determined endogenously as part of the solution, by integrating the optimal conditional $\pi^*(\hat{p}\mid p)$ over states.

Intuitively, $q(\hat{p})$ represents what the agent would report *without any state-specific information*—their unconditional reporting behavior averaged over all forecasting problems they face. An agent who frequently encounters base rates near 1/2 (because f is concentrated there) will have q concentrated near 1/2; deviating from this default to report an extreme probability requires acquiring more information, and hence incurs a higher cost.

Importantly, the endogenous nature of q has **no effect on the main results when c is small**. In the Gaussian approximation (Section 2.2), the conditional distribution $\pi^*(\hat{p}\mid p)$ is sharply peaked around $\hat{p} = p$ for small c. Near this peak, any smooth positive reference distribution q produces the same variance formula: expanding $q(\hat{p}) = q(p) + q'(p)(\hat{p}-p) + \cdots$, the leading term $q(p)$ is absorbed into the normalizing constant, and the first correction shifts the mean by $O(c)$ without affecting the first-order term in the small-c expansion of the variance. The specific form of q matters only for corrections of order c^2 and higher, and for exact tail probabilities.

1.4 Why KL divergence?

The KL divergence (equivalently, mutual information) is not an arbitrary choice of cost function. It is the unique cost functional (up to scaling) that satisfies three natural axioms for information-processing costs:

1. **Additivity over independent decisions.** If the agent faces two independent problems, the total cost of choosing strategies for both equals the sum of the individual costs.
2. **Monotonicity.** A more precise strategy (one that is further from q in terms of achievable payoffs) costs more.
3. **Invariance to sufficient statistics.** The cost depends only on the statistical relationship between the strategy and the default, not on how the action space is parameterized.

These properties are established in the information theory literature and underpin the use of mutual information in rational inattention models (see Sims, 2003, and the subsequent literature).

1.5 Solution: the Gibbs distribution

The optimization problem (1) has a well-known closed-form solution in the discrete case (Matějka & McKay, 2015, Theorem 1). We derive the continuous analogue here for completeness.

Proposition 1 (Optimal reporting distribution)**.** *The optimal conditional reporting distribution for* (1) *is:*

$$\pi^*(\hat{p} \mid p) = \frac{1}{Z(p)}\, q(\hat{p})\, \exp\!\left(\frac{V_p(\hat{p})}{c}\right), \tag{2}$$

where $Z(p) = \int_0^1 q(\hat{p}) \exp(V_p(\hat{p})/c)\, d\hat{p}$ *is the normalizing constant, and* $q(\hat{p}) = \int_0^1 f(p)\, \pi^*(\hat{p} \mid p)\, dp$ *is the endogenous marginal distribution of reports. Equation* (2) *and the definition of* q *must be solved simultaneously as a fixed-point problem.*

Proof. Write the objective (1) in full, substituting the definition of mutual information:

$$\mathcal{J}[\pi] = \int_0^1 f(p) \int_0^1 \pi(\hat{p} \mid p) \left[V_p(\hat{p}) - c \ln \frac{\pi(\hat{p} \mid p)}{q(\hat{p})} \right] d\hat{p}\, dp.$$

We maximize $\mathcal{J}$ over $\pi(\hat{p} \mid p)$ subject to the constraint $\int_0^1 \pi(\hat{p} \mid p)\, d\hat{p} = 1$ for each p. Treating $q(\hat{p})$ as fixed (we verify consistency below), introduce Lagrange multipliers $\lambda(p)$ for the normalization constraints and take the functional derivative with respect to $\pi(\hat{p} \mid p)$:

$$\frac{\delta}{\delta \pi(\hat{p} \mid p)} \left[\mathcal{J}[\pi] - \int_0^1 \lambda(p) \left(\int_0^1 \pi(\hat{p} \mid p)\, d\hat{p} - 1 \right) dp \right] = 0.$$

This gives the first-order condition:

$$f(p) \left[V_p(\hat{p}) - c \ln \frac{\pi(\hat{p} \mid p)}{q(\hat{p})} - c \right] = \lambda(p).$$

Solving for $\pi(\hat{p} \mid p)$:

$$\pi(\hat{p} \mid p) = q(\hat{p}) \exp\!\left(\frac{V_p(\hat{p}) - c - \lambda(p)/f(p)}{c} \right).$$

The terms that do not depend on $\hat{p}$ are absorbed into the normalizing constant $Z(p)$, giving (2). The second functional derivative is $-c\, f(p)/\pi(\hat{p} \mid p) < 0$, confirming that this is a maximum. Finally, the marginal $q(\hat{p}) = \int_0^1 f(p)\, \pi^*(\hat{p} \mid p)\, dp$ must be consistent with the q appearing in (2), which determines q as the fixed point of this self-consistency condition. □

This is the continuous analogue of the multinomial logit model: the probability of reporting $\hat{p}$ given state p is proportional to the unconditional reporting probability $q(\hat{p})$ scaled exponentially by the payoff $V_p(\hat{p})/c$. Reports with higher expected scores are exponentially more likely, with the sensitivity controlled by $1/c$.

Limiting behavior.

- As $c \to 0$: $\pi^*(\hat{p} \mid p)$ concentrates on $\arg\max_{\hat{p}} V_p(\hat{p}) = p$ (by strict properness). The agent acquires perfect information and reports truthfully.
- As $c \to \infty$: $\pi^*(\hat{p} \mid p) \to q_0(\hat{p})$, independent of p. Information is too expensive; the agent's report reveals nothing about the state.

Automatic support. Since $\pi^*(\hat{p} \mid p)$ is supported on $[0, 1]$, the reported probability automatically satisfies $\hat{p} \in [0, 1]$. There is no boundary/support problem—unlike the additive noise model, where $\hat{p} = p - \theta/G''(p)$ can fall outside $[0, 1]$.

2 Application to scoring rules

2.1 Setup

The agent faces a strictly proper scoring rule S with entropy function G. The true probability $p \in (0, 1)$ is unknown to the agent, who holds prior $f(p)$. By (2), the optimal conditional reporting distribution is:

$$\pi^*(\hat{p} \mid p) = \frac{q(\hat{p}) \exp(V_p(\hat{p})/c)}{Z(p)}, \tag{3}$$

where $V_p(\hat{p}) = p\, S(\hat{p}, 1) + (1 - p)\, S(\hat{p}, 0)$ has a unique maximum at $\hat{p} = p$ by strict properness, and $q(\hat{p})$ is the endogenous marginal (Section 1.3). The agent's actual report is a draw from $\pi^*(\hat{p} \mid p)$.

2.2 Gaussian approximation

We now show that for small information cost c, the optimal reporting distribution $\pi^*(\hat{p} \mid p)$ is well approximated by a Gaussian. The argument proceeds in three steps: (i) a Taylor expansion of the expected score V_p around its maximum, (ii) control of the resulting approximation error on the full domain $[0, 1]$, and (iii) moment computation under the approximate density.

Step 1: Taylor expansion of V_p. Recall that every strictly proper scoring rule admits a representation in terms of a convex function G (the *entropy function* or *expected score function*):

$$S(\hat{p}, 1) = G(\hat{p}) + G'(\hat{p})(1 - \hat{p}), \qquad S(\hat{p}, 0) = G(\hat{p}) - G'(\hat{p})\, \hat{p}.$$

The expected score when the true probability is p is then:

$$V_p(\hat{p}) = p\, S(\hat{p}, 1) + (1 - p)\, S(\hat{p}, 0) = G(\hat{p}) + G'(\hat{p})(p - \hat{p}).$$

Differentiating with respect to $\hat{p}$:

$$V_p'(\hat{p}) = G'(\hat{p}) + \frac{d}{d\hat{p}}\big[G'(\hat{p})(p - \hat{p})\big] = G''(\hat{p})(p - \hat{p}),$$

which confirms $V_p'(p) = 0$. Differentiating once more:

$$V_p''(\hat{p}) = G'''(\hat{p})(p - \hat{p}) - G''(\hat{p}).$$

Evaluating at $\hat{p} = p$, the first term vanishes and we obtain $V_p''(p) = -G''(p)$. Since V_p has a strict maximum at p, we have $G''(p) > 0$.

By Taylor's theorem with integral remainder, for $\delta = \hat{p} - p$:

$$V_p(p + \delta) = V_p(p) - \tfrac{1}{2}\, G''(p)\, \delta^2 + E(\delta), \tag{4}$$

where the remainder satisfies

$$E(\delta) = \int_0^1 (1 - t)\big[V_p''(p + t\delta) - V_p''(p)\big]\, \delta^2\, dt.$$

We now state the approximation result precisely.

Proposition 2 (Gaussian approximation)**.** *Let S be a strictly proper scoring rule whose entropy function G is C^3 on $(0, 1)$, and let $p \in (0, 1)$. Assume:*

(A1) *$G''(p) > 0$;*

(A2) *G''' is bounded on a neighborhood of p: there exist $\eta > 0$ and $M < \infty$ such that $|G'''(\hat{p})| \le M$ for all $\hat{p} \in (p - \eta, p + \eta) \cap (0, 1)$;*

(A3) *The endogenous marginal q is positive and C^2 on a neighborhood of p.*

Then as $c \to 0^+$:

$$\mathbb{E}[\hat{p} - p \mid p] = \frac{q'(p)}{q(p)} \cdot \frac{c}{G''(p)} + O(c^{3/2}), \tag{5}$$

$$\mathrm{Var}(\hat{p} - p \mid p) = \frac{c}{G''(p)} + O(c^{3/2}). \tag{6}$$

Moreover, the total variation distance between $\pi^*(\cdot \mid p)$ and the Gaussian $N\big(0,\, c/G''(p)\big)$ is $O(\sqrt{c})$ as $c \to 0^+$.

Proof. Write $a = G''(p)/(2c)$ and $\sigma^2 = c/G''(p) = 1/(2a)$.

Step 2: Domain truncation and tail control. Define the truncated domain $I_c = [p - c^{1/3}, p + c^{1/3}] \cap [0, 1]$. We show that the mass outside I_c is exponentially negligible.

By assumption (A2), for $|\delta| \le \eta$ the remainder in (4) satisfies $|E(\delta)| \le M|\delta|^3/6$. For $|\delta| > \eta$, the strict properness of S guarantees $V_p(p + \delta) < V_p(p)$; moreover, by continuity and compactness of $[0, 1]$, we have $V_p(p + \delta) \le V_p(p) - r$ for some $r = r(\eta, p) > 0$ whenever $|\delta| \ge \eta$.

For c small enough that $c^{1/3} < \eta$, consider the tail integral:

$$T_c \coloneqq \int_{[0,1]\setminus I_c} q(\hat{p})\, \exp\big(V_p(\hat{p})/c\big)\, d\hat{p}.$$

Splitting into $\{|\delta| \in [c^{1/3}, \eta]\}$ and $\{|\delta| > \eta\}$: on the first region, $V_p(\hat{p}) \le V_p(p) - \frac{1}{2}G''(p)\delta^2 + M|\delta|^3/6 \le V_p(p) - \frac{1}{4}G''(p)\delta^2$ for c small enough (since $|\delta| \le \eta$ and $M|\delta|/3 \le G''(p)/2$ when $\eta \le 3G''(p)/(2M)$). On the second region, $V_p(\hat{p}) \le V_p(p) - r$. Therefore:

$$T_c \le \bar{q}\left[\int_{|\delta| \ge c^{1/3}} \exp\Big(-\frac{G''(p)}{4c}\delta^2\Big)\, d\delta + \exp(-r/c)\right],$$

where $\bar{q} = \sup_{[0,1]} q$. The Gaussian tail integral satisfies $\int_{|\delta|\geq c^{1/3}} \exp(-a\delta^2/2)\, d\delta \leq 2\exp(-ac^{2/3}/2)/\sqrt{a}$ by Mill's inequality. Since $a = G''(p)/(4c)$, the exponent is $-G''(p)c^{-1/3}/8$, which diverges as $c \to 0$. Both terms are therefore $o(c^N)$ for every $N > 0$.

Meanwhile, the mass on I_c satisfies:

$$\int_{I_c} q(\hat{p})\, \exp(V_p(\hat{p})/c)\, d\hat{p} \geq \underline{q}\, \exp(V_p(p)/c) \int_{-\sigma}^{\sigma} \exp(-a\delta^2)\, d\delta,$$

where $\underline{q} = \inf_{I_c} q > 0$ for small c, and $\sigma = c^{1/3} \gg \sqrt{c/G''(p)}$. The inner integral is $\sqrt{\pi/a}\,(1-o(c^N))$. Thus $T_c/\int_{I_c} = o(c^N)$ for every N, and all moments of $\pi^*(\cdot \mid p)$ may be computed on I_c up to superpolynomially small error.

Step 3: Moment computation on I_c. On I_c, where $|\delta| \leq c^{1/3}$, the remainder satisfies $|E(\delta)| \leq Mc/6$ (since $|\delta|^3 \leq c$). Write:

$$\frac{V_p(p+\delta)}{c} = \frac{V_p(p)}{c} - a\delta^2 + \frac{E(\delta)}{c}.$$

Since $|E(\delta)/c| \leq M|\delta|^3/(6c)$ and $|\delta| \leq c^{1/3}$, we have $|E(\delta)/c| \leq M/(6c^0) = M/6$, but more precisely, for typical $\delta \sim \sqrt{c}$: $|E(\delta)/c| = O(c^{1/2})$. To make this rigorous, write:

$$\exp\big(E(\delta)/c\big) = 1 + E(\delta)/c + O\big((E(\delta)/c)^2\big).$$

On I_c, the remainder $E(\delta)/c$ admits the bound $|E(\delta)/c| \leq M|\delta|^3/(6c)$. Under the Gaussian weight $e^{-a\delta^2}$, the moments of $|\delta|^k$ are $O(c^{k/2})$, so $\langle |E(\delta)/c| \rangle_a = O(c^{1/2})$ and $\langle (E(\delta)/c)^2 \rangle_a = O(c)$.

The conditional density on I_c is therefore:

$$\pi^*(\hat{p} \mid p) \propto q(p+\delta)\, \big(1 + E(\delta)/c + O(c)\big)\, \exp(-a\delta^2). \tag{7}$$

Now expand q around p. By assumption (A3), $q(p+\delta) = q(p)\big[1+\beta\,\delta+O(\delta^2)\big]$ where $\beta = q'(p)/q(p)$. Combining with the remainder expansion:

$$\pi^*(\hat{p} \mid p) \propto \big(1 + \beta\,\delta + O(\delta^2) + O(|\delta|^3/c)\big)\, \exp(-a\delta^2).$$

We compute moments term by term, using the standard Gaussian integral identities on $\mathbb{R}$ (the extension from I_c to $\mathbb{R}$ introduces only superpolynomially small error by Step 2):

$$\int_{-\infty}^{\infty} e^{-a\delta^2}\, d\delta = \sqrt{\frac{\pi}{a}}, \quad \int_{-\infty}^{\infty} \delta^2\, e^{-a\delta^2}\, d\delta = \frac{1}{2a}\sqrt{\frac{\pi}{a}}, \quad \int_{-\infty}^{\infty} \delta\, e^{-a\delta^2}\, d\delta = 0,$$

the last by antisymmetry. All odd-power integrals $\int \delta^{2k+1} e^{-a\delta^2}\, d\delta = 0$.

Mean. The normalizing constant is $Z = \sqrt{\pi/a}\,\big(1 + O(c)\big)$. The first moment integral gives:

$$\int \delta\,(1+\beta\delta)\, e^{-a\delta^2}\, d\delta = \beta \cdot \frac{1}{2a}\sqrt{\frac{\pi}{a}},$$

where the $\int \delta\, e^{-a\delta^2}\, d\delta = 0$ term vanishes. The $O(\delta^2)$ and $O(|\delta|^3/c)$ perturbations contribute $O(c^{3/2})$ to the mean (the $O(\delta^2)$ term yields an odd integral $\int \delta \cdot \delta^2\, e^{-a\delta^2}\, d\delta = 0$; the cross-terms with $E(\delta)/c$ contribute at most $O(c^{1/2}) \cdot O(c) = O(c^{3/2})$ by Cauchy–Schwarz). Therefore:

$$\mathbb{E}[\delta] = \frac{\beta}{2a} + O(c^{3/2}) = \frac{q'(p)}{q(p)} \cdot \frac{c}{G''(p)} + O(c^{3/2}).$$

Second moment. Similarly:

$$\int \delta^2 \left(1 + \beta\delta\right) e^{-a\delta^2}\, d\delta = \frac{1}{2a}\sqrt{\frac{\pi}{a}} + 0,$$

where the $\beta \int \delta^3 e^{-a\delta^2}\, d\delta = 0$ by antisymmetry. The perturbation terms contribute $O(c^{3/2})$ to $\langle\delta^2\rangle$ by the same reasoning. Thus $\langle\delta^2\rangle = 1/(2a) + O(c^{3/2}) = c/G''(p) + O(c^{3/2})$.

Variance. Combining:

$$\mathrm{Var}(\delta) = \langle\delta^2\rangle - \langle\delta\rangle^2 = \frac{c}{G''(p)} + O(c^{3/2}) - O(c^2) = \frac{c}{G''(p)} + O(c^{3/2}).$$

Total variation bound. Let φ_σ denote the density of $N(0, \sigma^2)$ with $\sigma^2 = c/G''(p)$. On I_c, the density ratio $\pi^*(p + \delta \mid p)/\varphi_\sigma(\delta)$ equals $1 + \beta\delta + O(\delta^2) + O(|\delta|^3/c)$ times a normalizing correction of $1 + O(c)$. Under φ_σ, the expected absolute perturbation is $\langle|\beta\delta|\rangle_{\varphi_\sigma} = |\beta|\sqrt{2\sigma^2/\pi} = O(\sqrt{c})$. By Scheffé's identity, $d_{\mathrm{TV}}(\pi^*, \varphi_\sigma) = \frac{1}{2}\int |\pi^* - \varphi_\sigma| \le O(\sqrt{c})$. □

Remark 1. *Assumptions (A1)–(A2) are satisfied by all commonly used proper scoring rules. For the log score, $G(p) = -p\ln p - (1-p)\ln(1-p)$ gives $G''(p) = 1/[p(1-p)]$ and $G''' = (2p-1)/[p(1-p)]^2$, both bounded on any compact subset of $(0, 1)$. For the Brier score, $G(p) = p(1-p)$ gives $G''(p) = 2$ and $G''' = 0$. Assumption (A3) is a mild regularity condition on the fixed-point marginal q, which inherits smoothness from the Gibbs kernel.*

The key consequence is:

$$\hat{p} - p \;\sim\; N\!\left(0,\; \frac{c}{G''(p)}\right) + O_{\mathrm{TV}}(\sqrt{c}) \quad \text{as } c \to 0^+. \tag{8}$$

2.3 Reporting variance

The key result is:

$$\boxed{\mathrm{Var}(\hat{p} - p \mid p) \approx \frac{c}{G''(p)}.} \tag{9}$$

This should be compared with the additive noise model, where $\mathrm{Var}(\hat{p} - p) \approx \sigma^2/[G''(p)]^2$. The two models differ in two ways:

1. **Dependence on curvature.** The rational inattention variance scales as $1/G''(p)$, while the additive noise variance scales as $1/[G''(p)]^2$. The rational inattention model has a *weaker* dependence on curvature: doubling the curvature halves the variance (rather than reducing it by a factor of four). This is because in the RI model, the noise is endogenous—the agent's precision adapts to the payoff landscape—whereas in the additive noise model, the noise magnitude σ^2 is fixed and only the "restoring force" $G''(p)$ changes.

2. **Behavioral parameter.** The noise level is controlled by c (information cost) rather than σ^2 (exogenous noise variance). The parameter c has a clear interpretation: it is the marginal cost of information processing, measured in the same units as the scoring rule payments.

Remark 2 (Intuition for the difference in curvature dependence)**.** *In the additive noise model, the agent faces a fixed perturbation θ to their first-order condition and must "resist" it using the curvature $G''(p)$ of the expected score. The resulting error $\hat{p} - p \approx \theta/G''(p)$ depends on the ratio of noise to curvature, giving variance $\propto 1/[G''(p)]^2$.*

In the rational inattention model, the agent optimally chooses how much information to acquire. The Gibbs distribution concentrates where V_p is high relative to its surroundings; the "width" of this concentration is determined by the ratio $c/|V_p''(p)| = c/G''(p)$. Higher curvature $G''(p)$ means the payoff drops off more steeply around the truth, making it more valuable to acquire precise information—but the curvature enters the variance only once (through the width of the Gibbs peak), not twice as in the noise-to-curvature ratio of the additive model.

3 The principal's problem under rational inattention

3.1 The principal's decision problem

The principal faces a binary uncertain event $X \in \{0, 1\}$ with $\Pr(X = 1) = p$, and must choose a decision d to maximize expected utility $u(d, p) = p\, u(d, 1) + (1 - p)\, u(d, 0)$. The principal does not know p and delegates its estimation to the agent; given the agent's report $\hat{p}$, the principal acts with the optimal decision

$$d^*(\hat{p}) = \arg\max_d \; \hat{p}\, u(d, 1) + (1 - \hat{p})\, u(d, 0),$$

which satisfies the first-order condition

$$\hat{p}\, u'(d^*(\hat{p}), 1) + (1 - \hat{p})\, u'(d^*(\hat{p}), 0) = 0. \tag{10}$$

We assume $u(d, x)$ is strictly concave in d for each x and that $d^*(\hat{p})$ is uniquely defined with $(d^*)'(\hat{p}) \neq 0$.

3.2 Expected loss from reporting error

When the true probability is p but the agent reports $\hat{p}$, the principal's loss relative to the full-information benchmark is:

$$L = u(d^*(p), p) - u(d^*(\hat{p}), p).$$

We expand $u(d^*(\hat{p}), p)$ to second order in $\delta = \hat{p} - p$ by differentiating with respect to $\hat{p}$. The first derivative is:

$$\frac{d}{d\hat{p}}\, u(d^*(\hat{p}), p) = u'(d^*(\hat{p}), p) \cdot (d^*)'(\hat{p}).$$

At $\hat{p} = p$, this vanishes because $u'(d^*(p), p) = 0$ by the first-order condition (10). The second derivative is:

$$\frac{d^2}{d\hat{p}^2}\, u(d^*(\hat{p}), p) = u''(d^*(\hat{p}), p) \cdot [(d^*)'(\hat{p})]^2 + u'(d^*(\hat{p}), p) \cdot (d^*)''(\hat{p}).$$

At $\hat{p} = p$, the second term again vanishes ($u'(d^*(p), p) = 0$), leaving:

$$\left.\frac{d^2}{d\hat{p}^2}\, u(d^*(\hat{p}), p)\right|_{\hat{p}=p} = u''(d^*(p), p) \cdot [(d^*)'(p)]^2,$$

where $u''(d^*(p), p) = p\, u''(d^*(p), 1) + (1 - p)\, u''(d^*(p), 0) < 0$ by strict concavity. Define the *principal's sensitivity*:

$$H(p) \equiv u''(d^*(p), p) \cdot [(d^*)'(p)]^2. \tag{11}$$

Note that $H(p) < 0$. The magnitude $|H(p)|$ measures how sensitive the principal's utility is to small errors in the reported probability: it is large when the utility is sharply curved at the optimum

(large $|u''|$) or when the optimal decision changes rapidly with the estimate (large $|(d^*)'|$). The Taylor expansion gives:

$$L \approx -\frac{1}{2}\, H(p) \cdot \delta^2 = \frac{1}{2}\, |H(p)| \cdot \delta^2.$$

Taking expectations using the reporting variance (9):

$$\mathbb{E}[L \mid p] \approx \frac{|H(p)|\, c}{2\, G''(p)}. \tag{12}$$

Compare with the additive noise model: $\mathbb{E}[L \mid p] \approx \sigma^2 |H(p)|/[2\, G''(p)^2]$. The expected loss now scales as $1/G''(p)$ rather than $1/[G''(p)]^2$.

3.3 The scaling problem and net utility

Replacing S with aS for $a > 1$ scales $G''(p)$ by a, reducing the expected loss by a factor of a. As $a \to \infty$, the loss vanishes for any strictly proper scoring rule, so comparing rules without constraining the scale of payments is meaningless. We therefore introduce a cost $\alpha > 0$ per unit of expected payment: when the agent reports approximately truthfully, the expected payment at belief p is $G(p)$, and the principal's net expected utility at a given p is

$$-\mathbb{E}[L \mid p] - \alpha\, G(p) \approx -\frac{c\, |H(p)|}{2\, G''(p)} - \alpha\, G(p).$$

Integrating over the prior $f(p)$ on p, the principal's problem is:

$$\max_{S \in \mathcal{S}} \int_0^1 \left[-\frac{c\, |H(p)|\, f(p)}{2\, G''(p)} - \alpha\, G(p)\, f(p) \right] dp, \tag{13}$$

where $\mathcal{S}$ is the set of strictly proper scoring rules with $S(\hat{p}, x) \geq 0$ (limited liability).

3.4 Green's function representation

To optimize over the curvature $g(p) = G''(p) > 0$, we express G as a linear functional of g. The standard representation of proper scoring rules gives $S(\hat{p}, 1) = G(\hat{p}) + (1 - \hat{p})\, G'(\hat{p})$ and $S(\hat{p}, 0) = G(\hat{p}) - \hat{p}\, G'(\hat{p})$; limited liability $S(\hat{p}, x) \geq 0$ reduces to the two boundary conditions $G(0) + G'(0) \geq 0$ and $G(1) - G'(1) \geq 0$. Since adding an affine function to G does not change G'', the principal minimizes payments by choosing both constraints to bind:

$$G(0) + G'(0) = 0, \qquad G(1) - G'(1) = 0.$$

Integrating $G''(p) = g(p)$ twice and applying these boundary conditions, one obtains (see the derivation in the companion note):

$$G(p) = \int_0^1 K(p, t)\, g(t)\, dt, \tag{14}$$

where the Green's function is:

$$K(p, t) = \begin{cases} p(1 - t), & t \leq p, \\ t(1 - p), & t > p. \end{cases} \tag{15}$$

Note that $K(p, t) = K(t, p) \geq 0$, so $G(p) \geq 0$.

3.5 Separable reformulation

Substituting (14) into the payment term and swapping the order of integration, we define the *cost-of-curvature function*:

$$\phi(p) \equiv \int_0^1 K(t,p)\, f(t)\, dt. \tag{16}$$

The magnitude $\phi(p)$ measures the prior-weighted cost of adding curvature at p: increasing $g(p)$ raises $G(t)$ for all t, and $\phi(p)$ aggregates this increase weighted by f. The principal's objective (13) becomes:

$$\max_{g>0} \int_0^1 \left[-\frac{c\,|H(p)|\, f(p)}{2\, g(p)} - \alpha\, \phi(p)\, g(p) \right] dp. \tag{17}$$

This is separable in g: the integrand at each p depends only on $g(p)$, so the optimization can be performed pointwise.

3.6 Pointwise optimization

At each p, we maximize:

$$h(g) = -\frac{c\,|H(p)|\, f(p)}{2g} - \alpha\, \phi(p)\, g.$$

Both terms are negative (h is the sum of a term $\propto -1/g$ and a term $\propto -g$). The first derivative is:

$$h'(g) = \frac{c\,|H(p)|\, f(p)}{2g^2} - \alpha\, \phi(p).$$

Setting $h'(g) = 0$:

$$g^2 = \frac{c\,|H(p)|\, f(p)}{2\alpha\, \phi(p)}.$$

The second derivative is $h''(g) = -c\,|H(p)|\, f(p)/g^3 < 0$, confirming a maximum. The optimal curvature is:

$$\boxed{G''(p)^*_{\mathrm{RI}} = \sqrt{\frac{c\,|H(p)|\, f(p)}{2\alpha\, \phi(p)}}.} \tag{18}$$

3.7 Comparison with the additive noise model

The additive noise model yields $G''(p)^* = (\sigma^2 |H(p)| f(p)/(\alpha\, \phi(p)))^{1/3}$. Both models share the same *curvature allocation principle*: concentrate curvature where

- errors are costly ($|H(p)|$ large),
- the base rate is likely ($f(p)$ large),
- curvature is cheap ($\phi(p)$ small).

The key difference is the *exponent*: 1/2 under rational inattention versus 1/3 under additive noise. The RI model produces a more heterogeneous curvature profile—it responds more aggressively to variation in $|H(p)|\, f(p)/\phi(p)$. This is because the weaker dependence of variance on curvature ($1/g$ vs. $1/g^2$) means the marginal benefit of additional curvature diminishes less rapidly, so it pays to push curvature higher where the benefit-to-cost ratio is favorable.

Table 1: Comparison of the two models.

	Additive noise	Rational inattention
Reporting variance	$\sigma^2/[G''(p)]^2$	$c/G''(p)$
Expected loss	$\sigma^2\lvert H\rvert/[2g^2]$	$c\lvert H\rvert/[2g]$
Optimal curvature	$\left(\frac{\sigma^2\lvert H\rvert f}{\alpha\phi}\right)^{1/3}$	$\left(\frac{c\lvert H\rvert f}{2\alpha\phi}\right)^{1/2}$
Exponent	$1/3$	$1/2$
Noise parameter	σ^2 (exogenous)	c (information cost)
Support of $\hat{p}$	$\mathbb{R}$ (requires clipping)	$[0,1]$ (automatic)

4 When is S_u optimal?

For the decision-matched scoring rule S_u, the curvature is $G''(p) = |H(p)|$. Substituting into (18), S_u is optimal if and only if:

$$|H(p)| = \sqrt{\frac{c\,|H(p)|\,f(p)}{2\alpha\,\phi(p)}},$$

which simplifies to:

$$|H(p)|\,\phi(p)/f(p) = c/(2\alpha) \quad \text{(constant in } p\text{)}. \tag{19}$$

Counterexample. For $u(d,x) = -(d-x)^2$ and $f \equiv 1$: $|H(p)| = 2$ and $\phi(p) = (p^2 - p + 1)/2$. The condition (19) requires $2 \cdot (p^2 - p + 1)/2 = p^2 - p + 1$ to be constant, which fails (it ranges from 3/4 to 1). The Brier score is still suboptimal, but for the same reason as before: $\phi(p)$ is not constant.

The optimal curvature in this case is:

$$G''(p)^*_{\mathrm{RI}} = \sqrt{\frac{c \cdot 2}{\alpha(p^2 - p + 1)}} = \sqrt{\frac{2c}{\alpha(p^2 - p + 1)}}.$$

5 Existence of the optimal scoring rule

The pointwise optimization yields an optimal curvature function $g^*(p) = G''^*_{\mathrm{RI}}(p)$, but we must verify that a strictly proper scoring rule with this curvature actually exists. Specifically, we need to show that integrating g^* twice and imposing the boundary conditions produces a valid entropy function G^*.

By the Green's function representation (14), define

$$G^*(p) = \int_0^1 K(p,t)\,g^*(t)\,dt,$$

where K is the kernel (15). This integral is well-defined and finite for all $p \in [0,1]$ provided g^* is integrable on $(0,1)$, which holds whenever $|H(p)|\,f(p)/\phi(p)$ is bounded (guaranteed by $\phi(p) > 0$ on $(0,1)$ and $|H(p)|\,f(p)$ bounded).

We verify that G^* satisfies all requirements for a valid entropy function of a strictly proper scoring rule with limited liability.

1. **Strict convexity.** By construction, $(G^*)''(p) = g^*(p) > 0$ for all $p \in (0,1)$, so G^* is strictly convex.

2. **Boundary conditions.** Any G constructed via (14) automatically satisfies the boundary conditions $G(0)+G'(0) = 0$ and $G(1)-G'(1) = 0$. To see this: $G^*(0) = \int_0^1 K(0,t)\, g^*(t)\, dt = 0$ since $K(0,t) = 0$ for all t. Differentiating (14) gives $(G^*)'(0) = -\int_0^1 t\, g^*(t)\, dt$, and from the double integration $G^*(0) = \int_0^1 t\, g^*(t)\, dt$, so $G^*(0) + (G^*)'(0) = 0$. The second boundary condition follows analogously.

3. **Limited liability.** Since the boundary conditions bind with equality, the reduced limited liability constraints hold. The standard representation gives $\min_{\hat{p}} S(\hat{p}, 1) = G^*(0)+(G^*)'(0) = 0 \geq 0$ and $\min_{\hat{p}} S(\hat{p}, 0) = G^*(1) - (G^*)'(1) = 0 \geq 0$. Therefore $S(\hat{p}, x) \geq 0$ for all $\hat{p} \in (0,1)$ and $x \in \{0,1\}$.

4. **Non-negative expected payments.** Since $K(p,t) \geq 0$ and $g^*(t) > 0$, we have $G^*(p) \geq 0$ for all $p \in [0,1]$, with $G^*(p) > 0$ for $p \in (0,1)$.

The scoring rule corresponding to G^* is then given by the standard representation:

$$S^*(\hat{p}, 1) = G^*(\hat{p}) + (1-\hat{p})\,(G^*)'(\hat{p}), \qquad S^*(\hat{p}, 0) = G^*(\hat{p}) - \hat{p}\,(G^*)'(\hat{p}).$$

This is a strictly proper scoring rule satisfying limited liability, confirming that the optimal curvature (18) is attained by an admissible scoring rule.

Remark 3 (Uniqueness)**.** *The optimal scoring rule S^* is unique within the class of limited-liability strictly proper scoring rules. The objective* (17) *is strictly concave in $g(p)$ at each p (since $h''(g) < 0$), so the pointwise maximizer g^* is unique. The Green's function map $g \mapsto G$ is linear and injective, and the two boundary conditions pin down the affine degrees of freedom that are otherwise free for proper scoring rules. There is therefore no residual indeterminacy.*

Remark 4 (The optimal rule is generically novel)**.** *The optimal curvature $g^*(p) \propto \sqrt{|H(p)|\, f(p)/\phi(p)}$ depends on three problem-specific functions and does not, in general, belong to any standard parametric family of scoring rules. For instance, the power family $G''(p) \propto [p(1-p)]^{\beta-2}$ (which includes the Brier score at $\beta = 2$ and the logarithmic score as $\beta \to 1$) requires $|H(p)|\, f(p)/\phi(p) \propto [p(1-p)]^{2\beta-4}$, a condition that generically fails. Even in the simplest counterexample ($u(d,x) = -(d-x)^2$, $f \equiv 1$), the optimal curvature $g^*(p) \propto (p^2 - p + 1)^{-1/2}$ is not of this form. The optimal scoring rule is thus tailored to the specific decision problem and prior, rather than belonging to the standard toolkit of off-the-shelf rules.*

6 Tail probability of large errors

A major advantage of the rational inattention model is that the Gibbs distribution provides exponential bounds on the probability of large reporting errors, with the exponential rate determined exactly by the scoring rule's regret structure.

6.1 The regret function

For a strictly proper scoring rule S, the expected score $V_p(\hat{p})$ has a unique global maximum at $\hat{p} = p$. Define the *regret function*:

$$R_p(\hat{p}) \equiv V_p(p) - V_p(\hat{p}) \geq 0, \tag{20}$$

with equality only at $\hat{p} = p$. The regret $R_p(\hat{p})$ measures the expected score the agent sacrifices by reporting $\hat{p}$ instead of the truth. By strict properness, R_p is strictly positive for $\hat{p} \neq p$, continuous, and satisfies $R_p(\hat{p}) \to R_p(0) > 0$ and $R_p(\hat{p}) \to R_p(1) > 0$ as $\hat{p}$ approaches the boundaries.

6.2 Exponential tail bound

The tail probability of a reporting error exceeding $\Delta > 0$ is obtained by integrating the Gibbs distribution (3) over the error region:

$$\Pr(|\hat{p} - p| > \Delta \mid p) = \int_{|\hat{p}-p|>\Delta} \pi^*(\hat{p} \mid p)\, d\hat{p} = \int_{|\hat{p}-p|>\Delta} \frac{q(\hat{p})\, \exp(V_p(\hat{p})/c)}{Z(p)}\, d\hat{p}.$$

Since the normalizing constant is $Z(p) = \int_0^1 q(\hat{p})\, \exp(V_p(\hat{p})/c)\, d\hat{p}$, this becomes:

$$\Pr(|\hat{p} - p| > \Delta \mid p) = \frac{\int_{|\hat{p}-p|>\Delta} q(\hat{p})\, \exp(V_p(\hat{p})/c)\, d\hat{p}}{\int_0^1 q(\hat{p})\, \exp(V_p(\hat{p})/c)\, d\hat{p}}. \tag{21}$$

Define the *minimum regret at distance* Δ:

$$R^*(\Delta, p) \equiv \min_{\substack{\hat{p}\in[0,1] \\ |\hat{p}-p|\geq\Delta}} R_p(\hat{p}) = V_p(p) - \max_{\substack{\hat{p}\in[0,1] \\ |\hat{p}-p|\geq\Delta}} V_p(\hat{p}). \tag{22}$$

This is the smallest payoff sacrifice the agent incurs by making an error of at least Δ. By strict properness and continuity, $R^*(\Delta, p) > 0$ for all $\Delta > 0$ and $p \in (0, 1)$, and R^* is increasing in Δ (for $\Delta < \max(p, 1 - p)$).

Proposition 3 (Exponential tail bound)**.** *For any strictly proper scoring rule S, information cost $c > 0$, true probability $p \in (0, 1)$, and threshold $\Delta > 0$, the tail probability decays exponentially in the minimum regret:*

$$\Pr(|\hat{p} - p| > \Delta \mid p) = O\left(\exp\left(-\frac{R^*(\Delta, p)}{c}\right)\right) \quad \text{as } c \to 0, \tag{23}$$

where the implicit constant depends on p, Δ, and the marginal q, but not on c.

Proof. Let $\bar{q} = \sup_{\hat{p}\in[0,1]} q(\hat{p})$ and $\underline{q}(\epsilon) = \inf_{\hat{p}\in[p-\epsilon,\, p+\epsilon]} q(\hat{p})$ for some small $\epsilon > 0$. Since q is a positive continuous density on $(0, 1)$, both are finite and positive.

In the numerator of (21), for every $\hat{p}$ with $|\hat{p} - p| \geq \Delta$:

$$q(\hat{p})\, \exp(V_p(\hat{p})/c) \leq \bar{q}\, \exp\big((V_p(p) - R^*(\Delta, p))/c\big).$$

Therefore the numerator is at most $\bar{q}\, \exp\big((V_p(p) - R^*)/c\big)$.

The denominator is bounded below by restricting to a neighborhood of p. Choose $\epsilon \leq \min(p, 1 - p, \Delta)$. On $[p - \epsilon,\, p + \epsilon]$, we have $V_p(\hat{p}) \geq V_p(p) - R^*(\epsilon, p)$, so:

$$\int_0^1 q(\hat{p})\, \exp(V_p(\hat{p})/c)\, d\hat{p} \geq 2\epsilon\, \underline{q}(\epsilon)\, \exp\big((V_p(p) - R^*(\epsilon, p))/c\big).$$

Taking the ratio:

$$\Pr(|\hat{p} - p| > \Delta \mid p) \leq \frac{\bar{q}}{2\epsilon\, \underline{q}(\epsilon)}\, \exp\left(-\frac{R^*(\Delta, p) - R^*(\epsilon, p)}{c}\right).$$

The constant $\frac{\bar{q}}{2\epsilon\,\underline{q}(\epsilon)}$ depends on q and ϵ but not on c. Since $R^*(\epsilon, p) \to 0$ as $\epsilon \to 0$ while $R^*(\Delta, p) > 0$, we can choose ϵ small enough that $R^*(\Delta, p) - R^*(\epsilon, p) > 0$. The bound then has the form $C \cdot \exp(-(\text{positive constant})/c)$ where C does not depend on c, establishing the exponential decay (23). □

The bound (23) has several notable features:

- **Exponential rate from the Gibbs structure.** The rate $R^*(\Delta, p)/c$ is a direct consequence of the Gibbs form (2)—not of a Gaussian approximation. The exponential decay holds for the exact optimal reporting distribution at any c; only the multiplicative constant C depends on the endogenous marginal q.
- **Structural.** The exponential rate depends on the scoring rule only through the regret function R_p, which directly measures how strongly the scoring rule penalizes errors of a given size. Scoring rules with larger regret at distance Δ produce exponentially smaller tail probabilities.
- **Properness is sufficient.** The bound requires only that V_p has a strict global maximum at p—i.e., that the scoring rule is strictly proper. No additional regularity (smoothness, convexity of G, etc.) is needed.

6.3 Quadratic approximation of the regret

For small Δ, consider a report $\hat{p}$ with $|\hat{p} - p| = \Delta$. The regret is well approximated by the second-order Taylor expansion of V_p around p:

$$R_p(\hat{p}) = V_p(p) - V_p(\hat{p}) \approx \frac{1}{2}\, G''(p)\, (\hat{p} - p)^2 = \frac{1}{2}\, G''(p)\, \Delta^2, \tag{24}$$

where the first-order term vanishes because $V_p'(p) = 0$ by properness. Substituting into the tail bound (23):

$$\Pr(|\hat{p} - p| > \Delta \mid p) \;\lesssim\; \exp\!\left(-\frac{G''(p)\,\Delta^2}{2c}\right). \tag{25}$$

Note that the exponent $G''(p)\,\Delta^2/(2c)$ is exactly $\Delta^2/(2\sigma_p^2)$ where $\sigma_p^2 = c/G''(p)$ is the Gaussian approximation variance from (8). So the quadratic regret approximation recovers the same tail decay as the Gaussian approximation (8). The point of the Gibbs framework is that exponential tail decay holds more generally: it follows from the structure of (3) and properness alone, without relying on the Gaussian approximation.

6.4 Large errors and the exact regret

For large Δ—where the Gaussian approximation may be inaccurate—the exact regret $R^*(\Delta, p)$ provides a tighter bound. The regret function $R_p(\hat{p})$ encodes the full nonlinear shape of the expected score, not just its local curvature. In particular:

- For the **Brier score**, $V_p(\hat{p}) = -(\hat{p} - p)^2 + \text{const}$, so $R_p(\hat{p}) = (\hat{p} - p)^2$ and $R^*(\Delta, p) = \Delta^2$ (for $\Delta \le \max(p, 1-p)$). The quadratic approximation is exact.
- For the **log score**, $R_p(\hat{p}) = p \ln(p/\hat{p}) + (1-p)\ln((1-p)/(1-\hat{p}))$, the KL divergence between $(p, 1-p)$ and $(\hat{p}, 1-\hat{p})$. This grows faster than quadratically as $\hat{p} \to 0$ or $\hat{p} \to 1$, providing stronger tail protection near the boundaries than the Gaussian approximation would suggest.

6.5 Implications for scoring rule comparison

For two scoring rules S_1 and S_2 with regret functions $R_1^*(\Delta, p)$ and $R_2^*(\Delta, p)$, the ratio of tail probabilities is bounded by:

$$\frac{\Pr_1(|\hat{p} - p| > \Delta \mid p)}{\Pr_2(|\hat{p} - p| > \Delta \mid p)} \lesssim \exp\left(-\frac{R_1^*(\Delta, p) - R_2^*(\Delta, p)}{c}\right). \tag{26}$$

When $R_1^* > R_2^*$, scoring rule S_1 provides exponentially better tail protection, and the advantage grows as c decreases (more attentive agents) or as Δ increases (larger error thresholds). This exponential sensitivity to the regret structure makes the choice of scoring rule far more consequential for tail risk than the variance analysis alone would indicate.

7 Efficiency of the pass-through rule

7.1 The subcontracting setting

Suppose the principal is herself a forecaster, scored by a strictly proper scoring rule S_0 with entropy function G_0. She delegates the estimation of p to a sub-agent, whom she pays according to a scoring rule S of her choice. Her utility function is $u(d, x) = S_0(d, x)$, where d is the report she submits to the meta-principal. Since S_0 is strictly proper, the principal's optimal decision given the agent's report is simply $d^*(\hat{p}) = \hat{p}$, so $(d^*)'(\hat{p}) = 1$.

The principal's sensitivity (11) then simplifies to $H(p) = u''(d^*(p), p) = -G_0''(p)$, so $|H(p)| = G_0''(p)$.

A natural benchmark is the *pass-through rule*: the principal assigns the sub-agent a scaled version of her own rule S_0. Scaling S_0 by $a > 0$ scales the curvature by a, so the pass-through rule has $g(p) = a\, G_0''(p)$ for some $a > 0$ chosen optimally. We compare the principal's net utility under this rule against the pointwise-optimal curvature (18).

7.2 Optimal rule value

The principal's net utility under curvature profile g is given by the objective (17):

$$V(g) = \int_0^1 \left[-\frac{c\,|H(p)|\,f(p)}{2\,g(p)} - \alpha\,\phi(p)\,g(p)\right] dp.$$

In the subcontracting setting $|H(p)| = G_0''(p)$, so substituting the pointwise-optimal curvature $g^*(p) = \sqrt{c\,G_0''(p)\,f(p)/(2\alpha\,\phi(p))}$ from (18) into the integrand (writing $A = c\,G_0''\,f/2$ and $C = \alpha\phi$):

$$-\frac{A}{g^*} - C\,g^* = -\frac{A}{\sqrt{A/C}} - C\sqrt{A/C} = -2\sqrt{AC}.$$

Therefore:

$$V_{\mathrm{RI}}^* = -2\int_0^1 \sqrt{\frac{c\,G_0''(p)\,f(p)}{2} \cdot \alpha\,\phi(p)}\; dp = -\sqrt{2c\alpha}\int_0^1 \sqrt{G_0''(p)\,f(p)\,\phi(p)}\; dp. \tag{27}$$

7.3 Pass-through rule value

Under the pass-through rule, $g(p) = a\,G_0''(p)$. Substituting into the objective:

$$V(a) = \int_0^1 \left[-\frac{c\,G_0''\,f}{2a\,G_0''} - \alpha\,\phi\,a\,G_0''\right] dp = -\frac{c}{2a}\underbrace{\int_0^1 f(p)\,dp}_{=1} - \alpha\,a\int_0^1 \phi(p)\,G_0''(p)\,dp.$$

(Here $|H| = G_0''$ cancels in the first term.) Differentiating with respect to a and setting to zero:

$$V'(a) = \frac{c}{2a^2} - \alpha \int_0^1 \phi\, G_0''\, dp = 0 \qquad \Longrightarrow \qquad a^* = \sqrt{\frac{c}{2\alpha \int_0^1 \phi\, G_0''\, dp}}.$$

Substituting back:

$$V_{\text{pass}} = V(a^*) = -\sqrt{2c\alpha \int_0^1 \phi(p)\, G_0''(p)\, dp}.$$

7.4 Efficiency ratio

Since both V_{RI}^* and V_{pass} are negative, with V_{RI}^* closer to zero, define the ratio

$$\eta_{\text{RI}} = \frac{V_{\text{RI}}^*}{V_{\text{pass}}} = \frac{\displaystyle\int_0^1 \sqrt{G_0''(p)\, f(p)\, \phi(p)}\, dp}{\sqrt{\displaystyle\int_0^1 \phi(p)\, G_0''(p)\, dp}}, \tag{28}$$

where we used $\int f = 1$. We now show that $\eta \in (0, 1]$. Positivity is immediate since $G_0'', f, \phi > 0$. For the upper bound, we need:

$$\left(\int_0^1 \sqrt{G_0''\, f\, \phi}\, dp\right)^2 \leq \int_0^1 G_0''\, \phi\, dp.$$

Write $\int \sqrt{G_0''\, f\, \phi}\, dp = \int \sqrt{f} \cdot \sqrt{G_0''\, \phi}\, dp$ and apply the Cauchy–Schwarz inequality with $u = \sqrt{f}$ and $v = \sqrt{G_0''\, \phi}$:

$$\int \sqrt{G_0''\, f\, \phi}\, dp \leq \sqrt{\int f\, dp} \cdot \sqrt{\int G_0''\, \phi\, dp} = \sqrt{\int G_0''\, \phi\, dp}.$$

Squaring yields the desired inequality, so $\eta \leq 1$. Equality holds if and only if $\sqrt{f} \propto \sqrt{G_0''\, \phi}$, i.e., $G_0''(p)\, \phi(p)/f(p) = \text{const}$, which is exactly the condition (19) for S_u to be optimal. The fractional improvement from using the optimal rule is $1 - \eta$.

The efficiency ratio η is **parameter-free**: it depends only on the shape functions G_0'', f, and ϕ, not on c or α. The percentage gain from optimization is the same regardless of the attention cost or the cost of payments.

8 Extension to continuous outcomes

The preceding analysis assumes a binary event $X \in \{0, 1\}$, where the agent reports a single probability $\hat{p} = \Pr(X = 1)$. Many forecasting problems involve continuous outcomes: predicting a temperature, a stock return, or an unemployment rate. We now extend the rational inattention framework to this setting, focusing on the elicitation of the mean. The main finding is that the entire framework—Gibbs solution, Gaussian approximation, variance formula, optimal curvature, and curvature allocation principle—carries over with essentially no change. The binary case is a special case of the continuous one.

8.1 Consistent scoring functions for the mean

Let Y be a continuous random variable with support $[a, b]$ and mean $\mu = \mathbb{E}[Y]$, where $\mu \in [a, b]$. A principal wishes to elicit the mean from an agent. The agent reports $\hat{\mu} \in [a, b]$ and receives a payment $S(\hat{\mu}, y)$ when the outcome y is realized. The scoring function S is *strictly consistent for the mean* if the agent's expected score $V_\mu(\hat{\mu}) \equiv \mathbb{E}[S(\hat{\mu}, Y) \mid \mu]$ is uniquely maximized at $\hat{\mu} = \mu$ for every distribution on $[a, b]$ with mean μ.

Savage (1971) and Gneiting (2011) characterize such scoring functions. A scoring function $S(\hat{\mu}, y)$ is strictly consistent for the mean (over all distributions on $[a, b]$) if and only if it takes the form

$$S(\hat{\mu}, y) = G(\hat{\mu}) + G'(\hat{\mu})(y - \hat{\mu}) + h(y), \tag{29}$$

where $G : [a, b] \to \mathbb{R}$ is a strictly convex function (the *entropy function*), G' denotes its derivative, and $h : [a, b] \to \mathbb{R}$ is an arbitrary function of the outcome alone. The expected score is then:

$$V_\mu(\hat{\mu}) = G(\hat{\mu}) + G'(\hat{\mu})(\mu - \hat{\mu}) + \mathbb{E}[h(Y)]. \tag{30}$$

Since $\mathbb{E}[h(Y)]$ does not depend on the report $\hat{\mu}$, it is irrelevant for the agent's optimization and can be absorbed into a constant. The remaining terms $G(\hat{\mu}) + G'(\hat{\mu})(\mu - \hat{\mu})$ are **identical in form** to the binary case, with μ replacing p.

Differentiating (30) with respect to $\hat{\mu}$:

$$V'_\mu(\hat{\mu}) = G''(\hat{\mu})(\mu - \hat{\mu}), \tag{31}$$

which confirms $V'_\mu(\mu) = 0$ and

$$V''_\mu(\mu) = -G''(\mu). \tag{32}$$

The expected score has a strict maximum at $\hat{\mu} = \mu$ with curvature determined by $G''(\mu) > 0$, exactly as in the binary case.

Remark 5 (The binary case as a special case)**.** *When $Y \in \{0, 1\}$ is binary with $\Pr(Y = 1) = p$, the mean is $\mu = p$. A scoring function $S(\hat{\mu}, y)$ consistent for the mean of a binary variable is exactly a proper scoring rule $S(\hat{p}, x)$. Representation* (29) *reduces to:*

$$S(\hat{p}, 1) = G(\hat{p}) + G'(\hat{p})(1 - \hat{p}) + h(1), \qquad S(\hat{p}, 0) = G(\hat{p}) - G'(\hat{p})\hat{p} + h(0),$$

which is the standard entropy function representation of proper scoring rules (with the constants $h(1)$ and $h(0)$ absorbed by convention). The entire binary framework is a special case of mean elicitation with $[a, b] = [0, 1]$ and $Y \in \{0, 1\}$.

Remark 6 (Examples of consistent scoring functions for the mean)**.** *The* squared error *score $S(\hat{\mu}, y) = -(y - \hat{\mu})^2$ corresponds to $G(\hat{\mu}) = \hat{\mu}^2$, $h(y) = -y^2$, giving constant curvature $G''(\hat{\mu}) = 2$— the direct analogue of the Brier score.*

8.2 Transfer of results to mean elicitation

Since the expected score (30) has the same functional form $G(\hat{\mu}) + G'(\hat{\mu})(\mu - \hat{\mu})$ as in the binary case, every result from the preceding sections carries over with μ replacing p and $[a, b]$ replacing $[0, 1]$: the Gibbs solution (Proposition 1), the Gaussian approximation with variance $c/G''(\mu)$, the exponential tail bound (Proposition 3), and the optimal curvature formula (18). The efficiency analysis of the pass-through rule likewise applies verbatim with $G''_0(\mu)$ replacing $G''_0(p)$.

The only changes are mechanical. Under limited liability with $h(y) = 0$, the boundary conditions become:

$$G(a) + G'(a)(b-a) = 0, \qquad G(b) - G'(b)(b-a) = 0, \tag{33}$$

which for $[a, b] = [0, 1]$ reduce to the binary conditions $G(0) + G'(0) = 0$ and $G(1) - G'(1) = 0$. The Green's function for integrating $G''(\mu) = g(\mu)$ subject to (33) is:

$$K_{[a,b]}(\mu, t) = \frac{1}{b-a} \begin{cases} (\mu - a)(b - t), & t \le \mu, \\ (t - a)(b - \mu), & t > \mu, \end{cases} \tag{34}$$

so that $G(\mu) = \int_a^b K_{[a,b]}(\mu, t)\, g(t)\, dt$, generalizing (14). The optimal curvature is then:

$$G''(\mu)^* = \sqrt{\frac{c\,|H(\mu)|\, f(\mu)}{2\alpha\, \phi(\mu)}}, \tag{35}$$

where $\phi(\mu) = \int_a^b K_{[a,b]}(t, \mu)\, f(t)\, dt$, exactly as before.

8.3 Beyond the mean: quantile elicitation

The mean is not the only functional of interest. In risk management and distributional forecasting, quantiles are often the primary target. The rational inattention framework applies, but the structure differs in important ways.

Let F denote the CDF of Y and f_Y its density. For the τ-quantile q_τ (where $F(q_\tau) = \tau$), Gneiting (2011) shows that for any strictly increasing function g, the negatively oriented scoring function (loss)

$$L(\hat{q}, y) = (\mathbf{1}\{y \le \hat{q}\} - \tau)\big(g(\hat{q}) - g(y)\big)$$

is strictly consistent: the expected loss $\mathbb{E}[L(\hat{q}, Y)]$ is minimized at $\hat{q} = q_\tau$. To align with the rest of the paper (where the agent maximizes a score), we work with the negated score $S(\hat{q}, y) = -L(\hat{q}, y)$. The expected score is:

$$V_{q_\tau}(\hat{q}) = -\mathbb{E}\big[(\mathbf{1}\{Y \le \hat{q}\} - \tau)(g(\hat{q}) - g(Y))\big],$$

Writing out the expectation and splitting by whether $y \le \hat{q}$:

$$V(\hat{q}) = -\int_a^{\hat{q}} (g(\hat{q}) - g(y))\, f_Y(y)\, dy + \tau\big(g(\hat{q}) - \mathbb{E}[g(Y)]\big).$$

Differentiating with respect to $\hat{q}$ using the Leibniz rule $\frac{d}{d\hat{q}} \int_a^{\hat{q}} h(\hat{q}, y)\, dy = h(\hat{q}, \hat{q}) + \int_a^{\hat{q}} \frac{\partial}{\partial \hat{q}} h(\hat{q}, y)\, dy$ on the first term:

$$\frac{d}{d\hat{q}} \int_a^{\hat{q}} (g(\hat{q}) - g(y))\, f_Y(y)\, dy = \underbrace{(g(\hat{q}) - g(\hat{q}))\, f_Y(\hat{q})}_{=0} + \int_a^{\hat{q}} g'(\hat{q})\, f_Y(y)\, dy = g'(\hat{q})\, F(\hat{q}).$$

Combining with the second term:

$$V'(\hat{q}) = -g'(\hat{q})\, F(\hat{q}) + \tau\, g'(\hat{q}) = -(F(\hat{q}) - \tau)\, g'(\hat{q}),$$

confirming $V'(q_\tau) = 0$ since $F(q_\tau) = \tau$. Differentiating again:

$$V''(\hat{q}) = -f_Y(\hat{q})\, g'(\hat{q}) - (F(\hat{q}) - \tau)\, g''(\hat{q}).$$

At $\hat{q} = q_\tau$ the second term vanishes, giving:

$$V''(q_\tau) = -f_Y(q_\tau)\, g'(q_\tau), \tag{36}$$

which is strictly negative (confirming that the truth is a strict maximum). The reporting variance under rational inattention for quantile forecasts is therefore:

$$\mathrm{Var}(\hat{q} - q_\tau \mid q_\tau) \approx \frac{c}{f_Y(q_\tau)\, g'(q_\tau)}. \tag{37}$$

This differs from the mean case in two respects. First, the variance depends on the *density at the quantile* $f_Y(q_\tau)$—quantiles in regions where the density is high are easier to estimate precisely, because even a small shift in the report crosses many probability mass units. Second, the principal chooses the increasing function g, with g' playing the role of curvature. Increasing g' at a given quantile value sharpens incentives there, analogously to increasing G'' in the mean case.

The principal's optimization is more complex than in the mean case: the expected payment $\mathbb{E}[S(q_\tau, Y)]$ depends on g through the full distribution of Y (not just the quantile being elicited), so the separability that made the mean case tractable does not carry over directly. We leave the full analysis of optimal scoring functions for quantile elicitation as an open direction for future work.

9 Approximation by parametric families

The optimal scoring rule S^* is tailored to the specific decision problem u, prior f, and cost parameters c and α, and does not generally belong to any standard parametric family (Remark 4). In practice, a forecasting platform may prefer to deploy a rule from a simple parametric class. This section quantifies the welfare cost of such restrictions and identifies when simple rules suffice.

9.1 Best constant curvature (scaled Brier)

The simplest restriction is $g(p) = g_0$ (constant curvature), corresponding to a scaled Brier score. Substituting into the objective (17):

$$V(g_0) = -\frac{c\,\bar{H}}{2\,g_0} - \alpha\,\bar{\phi}\,g_0,$$

where $\bar{H} = \int_0^1 |H(p)|\, f(p)\, dp$ and $\bar{\phi} = \int_0^1 \phi(p)\, dp$. Optimizing over g_0 yields:

$$g_0^* = \sqrt{\frac{c\,\bar{H}}{2\alpha\,\bar{\phi}}}, \qquad V^*_{\text{Brier}} = -2\sqrt{\frac{c\,\alpha\,\bar{H}\,\bar{\phi}}{2}}. \tag{38}$$

Since both values are negative (with V^* closer to zero), we define the efficiency ratio as in Section 7.4:

$$\eta_{\text{Brier}} = \frac{V^*}{V^*_{\text{Brier}}} = \frac{\int_0^1 \sqrt{|H(p)|\, f(p)\, \phi(p)}\, dp}{\sqrt{\bar{H}\,\bar{\phi}}}. \tag{39}$$

By Cauchy–Schwarz (with $u = \sqrt{|H|f}$ and $v = \sqrt{\phi}$), $\eta_{\text{Brier}} \in (0, 1]$, with equality if and only if $|H(p)|\, f(p)/\phi(p)$ is constant in p—exactly the condition for the Brier score to be optimal. This ratio is parameter-free: it depends on the decision problem and prior, but not on c or α.

9.2 Best power-family rule

A richer one-parameter family is the *power class*: $g(p) = A\,[p(1-p)]^{-\gamma}$ for $\gamma \in [0,1)$ and $A > 0$. This nests the Brier score ($\gamma = 0$) and approaches the logarithmic score as $\gamma \to 1$. For each γ, substituting into (17) and optimizing over A gives:

$$A^*(\gamma) = \sqrt{\frac{c\int_0^1 |H(p)|\, f(p)\,[p(1-p)]^{\gamma}\,dp}{2\alpha\int_0^1 \phi(p)\,[p(1-p)]^{-\gamma}\,dp}},$$

and the corresponding net utility:

$$V^*_{\text{power}}(\gamma) = -2\sqrt{\frac{c\,\alpha}{2}}\sqrt{\int_0^1 |H(p)|\, f(p)\,[p(1-p)]^{\gamma}\,dp \;\cdot\; \int_0^1 \phi(p)\,[p(1-p)]^{-\gamma}\,dp}. \tag{40}$$

The optimal γ^* minimizes $V^*_{\text{power}}(\gamma)$ in magnitude, i.e., minimizes the product of the two integrals. This is a one-dimensional problem that can be solved numerically.

9.3 Examples

We evaluate the efficiency ratios for the counterexample $u(d,x) = -(d-x)^2$ (so $|H(p)| = 2$) under different priors.

Uniform prior. With $f \equiv 1$, $\phi(p) = (p^2 - p + 1)/2$. The averages are $\bar{H} = 2$ and $\bar{\phi} = 5/12$. Numerically, $\int_0^1 \sqrt{p^2 - p + 1}\,dp \approx 0.9120$, giving:

$$\eta_{\text{Brier}} = \frac{0.9120}{\sqrt{5/6}} \approx 0.999.$$

The welfare loss from using the Brier score is approximately 0.1%—negligible. This is because $\phi(p) = (p^2 - p + 1)/2$ varies only between 3/8 and 1/2 under the uniform prior, so the ratio $|H|\,f/\phi$ is nearly constant and the Brier score is nearly optimal. The best power-family rule ($\gamma^* \approx 0$) offers no meaningful improvement.

Concentrated priors. The approximation gap grows substantially when the prior concentrates probability mass unevenly. With a Beta$(2,2)$ prior ($f(p) = 6p(1-p)$, peaked at 1/2), the Brier efficiency drops to $\eta_{\text{Brier}} \approx 0.95$, a 5% welfare loss. With a Beta$(5,5)$ prior (more sharply peaked), $\eta_{\text{Brier}} \approx 0.79$—a 21% welfare loss. Asymmetric priors such as Beta$(2,5)$ yield similar gaps ($\eta_{\text{Brier}} \approx 0.73$).

The power family $g(p) = A[p(1-p)]^{-\gamma}$ provides little additional improvement in these cases: the best power rule achieves essentially the same efficiency as the Brier score. This is because the variation in $|H|\,f/\phi$ is driven by the interaction of the prior f with the cost-of-curvature function $\phi(p) = \int K(t,p)\,f(t)\,dt$, which depends on the prior in a way that $[p(1-p)]^{-\gamma}$ cannot approximate. The notable exception is the Beta$(1/2,1/2)$ prior, where $f(p) \propto [p(1-p)]^{-1/2}$ and the power family achieves $\eta_{\text{power}} \approx 1.00$—the optimal curvature itself lies (approximately) in the power class.

When does the gap matter? The efficiency ratio η_{Brier} is determined by the variation of $|H(p)|\,f(p)/\phi(p)$ across p. When this ratio is nearly constant—as with the uniform prior and symmetric loss—the Brier score is effectively optimal and the choice of scoring rule is inconsequential beyond properness. When the ratio varies substantially—due to concentrated or asymmetric priors, or non-constant $|H(p)|$ from asymmetric loss functions—the full optimal rule provides a meaningful welfare improvement that simple parametric families cannot capture.

10 Discussion

10.1 Summary of advantages

The rational inattention model offers several structural improvements over the additive noise approach:

1. **Principled micro-foundation.** The reporting distribution (Gibbs/Boltzmann) is derived from optimal information acquisition under cognitive constraints, not assumed as an exogenous noise structure. The functional form requires no arbitrary specification of a noise function $m(\hat{p})$.

2. **No boundary problem.** The Gibbs distribution on $[0, 1]$ automatically produces reports in $[0, 1]$. The clipping analysis required by the additive noise model is entirely unnecessary.

3. **Exact tail bounds.** The exponential tail decay at rate $R^*(\Delta, p)/c$ follows from the Gibbs structure and properness alone (Proposition 3), requiring no Gaussian approximation or Mill's ratio. The regret function R^* provides a complete, nonlinear characterization of tail risk that goes beyond the local curvature $G''(p)$.

4. **Interpretable parameter.** The information cost c has a clear meaning (marginal cost of acquiring information about the state) and is measured in the same units as the scoring rule payments, making it directly comparable to the payment cost α.

5. **Endogenous reference distribution.** The marginal $q(\hat{p})$ is determined as part of the solution, not specified by the modeler.

10.2 Potential concerns

1. **Is c exogenous?** The parameter c represents the agent's marginal cost of processing information about the state p. If the agent can choose their level of attention (e.g., by deciding how long to investigate), then c may itself depend on the scoring rule—agents may investigate more when the stakes are higher. This would lead to an endogenous c model with a two-stage structure: the principal designs the scoring rule, and the agent jointly chooses investigation effort and report. For now, we treat c as a fixed characteristic of the agent. This is the standard assumption in the rational inattention literature and is analogous to treating risk aversion as a fixed parameter in mechanism design.

2. **Higher-order corrections.** The Gaussian approximation (8) is a second-order expansion valid for small c. For large c or near the boundaries of $[0, 1]$, higher-order terms become important. However, unlike the additive noise model, the exact Gibbs distribution (3) remains available in closed form and can be evaluated numerically at any c. The tail bound of Proposition 3 holds without any small-c assumption.

3. **Endogenous marginal q.** As discussed in Section 1.3, the marginal distribution of reports $q(\hat{p})$ is determined endogenously as part of the solution, not specified by the modeler. This is a strength of the information acquisition formulation: it avoids the need to choose an arbitrary reference distribution. Moreover, as shown in Section 2.2, q does not affect the first-order term in the small-c expansion of the variance or optimal curvature; it enters only through the prefactor in the tail bound and corrections of order c^2 and higher.

10.3 Relationship to the additive noise model

An alternative model assumes the agent maximizes $V_p(\hat{p}) + \theta\hat{p}$ where $\theta \sim N(0, \sigma^2)$ is an exogenous noise term. This yields reporting variance $\sigma^2/[G''(p)]^2$ and optimal curvature with exponent 1/3 instead of 1/2. The two models share the same qualitative curvature allocation principle—concentrate curvature where errors are costly, base rates likely, and curvature cheap—but differ in their quantitative predictions.

The rational inattention model is the preferred framework for several reasons: it derives rather than assumes the noise structure, it avoids the boundary/clipping problem, and it provides exact (not approximate) tail bounds. Whether reporting variance scales as $1/G''$ or $1/[G'']^2$ is in principle an empirical question, testable with data on forecast dispersion under different scoring rules. The fact that both micro-foundations yield the same qualitative conclusions about optimal curvature allocation strengthens the paper's central message: it is not an artifact of a particular noise specification.